Converting Length Across Systems; Currency

Teacher's Edition

The Period

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Must Convert a length between customary and metric with a printed reference-sheet ratio, either direction, in a chain; convert money with a given rate, both ways. (MA.7.AR.3.3.)

Must not A ratio not on the FAST reference sheet; a real exchange rate the student must know; mass, volume or area (5.08–09).

Target I can convert a length between the customary and metric systems with a ratio from the reference sheet, and convert money with a given rate.

MTR **MTR.6.1** — Notes II and board 6: the estimate before the arithmetic · **MTR.4.1** — boards 4 and 6 split; the sixty-second re-vote on which ratio · **MTR.3.1** — boards 5–6: two conversions in a row, each one a form of 1, with the units cancelling on the paper

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	10 min
Example 1	8–9	4 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. A conversion ratio is a form of 1: multiply by it and only the unit changes.
2. Pick the ratio whose 1 is the unit you have. Estimate first — which unit is bigger?
3. Money is the same move with a rate someone sets, printed on the problem.

Slide by Slide

1. **Converting Length Across Systems; Currency.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the move: a ratio, and a division because feet are bigger. *1 retrieves 5.04 — real ÷ rate (last class). 2 retrieves 5.01 — copy ÷ original; k under 1 (last week). 3 retrieves 4.02 — decimal quotients (last unit). 4 retrieves MA.6.AR.3.5 — converting inside one system; today the ratio crosses systems (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 2.5 in 2. k = ½ 3. 5 4. 2.5 ft
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the ratio and the line.** 1 in = 2.54 cm, so 2.54 cm over 1 in is 1 — the top and bottom are the same length in two units.
6. **Notes.** The sheet gives both directions: 1 in = 2.54 cm and 1 cm = 0.39 in.
7. **Notes.** An exchange rate is a conversion ratio with a date on it: 0.92 euros per dollar today, something else tomorrow, so it is printed on the problem and never on the sheet.
8. **Example 1 — Inches to centimeters.** Ask which unit should give the bigger number before anyone multiplies.
9. **Example 1 — Worked.** 38.1 cm.
10. **Your Turn — Same steps, your numbers. Boards up when done.** 63.5 cm.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Dollars to euros, and back.** Ask what the second question's operation is before anyone writes it.
13. **Example 2 — Worked.** 73.60 euros; $50.
14. **Your Turn — Same steps, your numbers. Boards up when done.** 162 CAD; $60.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. 1 in = 2.54 cm. A ruler is 20 in long. How many centimeters is that? → ***50.8 cm*** *7.87 — divided by 2.54 [Notes II: centimeters are smaller, so more of them]; 22.54 — added the ratio*

18–19. Board 2. 1 cm = 0.39 in. A phone is 20 cm long. How many inches is that? → ***7.8 in*** *50.8 — multiplied by 2.54 [Notes II: pick the ratio whose 1 is the unit you have]; 0.78 — place value*

20–21. Board 3. 1 mi = 1.61 km. A race is 5 mi long. How many kilometers is that? → ***8.05 km*** *3.1 — multiplied by 0.62, the other ratio [Notes II]; 6.61 — added 1.61*

22–23. Board 4. 1 km = 0.62 mi. A trail is 5 km long. How many miles is that? → ***D — 3.1 mi*** *(A) 5 + 0.62: the additive error [7.AR.3.3: Math Nation's named misconception]; (B) place value: 5 × 0.62 is 3.1, not 0.31 [4.03]; (C) × 1.61, the ratio for the other direction [B1G-M 7.AR.3.3: the inverted ratio; Notes II] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

24–25. Board 5. 1 ft = 12 in and 1 in = 2.54 cm. A door is 6 ft tall. How many centimeters is that? → ***182.88 cm*** *15.24 — 6 × 2.54, the feet treated as inches*

[Notes I: the unit you have must match the ratio's 1]; 183 — rounded early (accept with the work shown)

26–27. Board 6. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. A table is 3 yd long. About how many centimeters is that? Estimate, then compute. → ***About 270; exactly 274.32 cm*** *7.62 — 3 × 2.54, the yards treated as inches [Notes II: estimate first — a 3-yd table is not 8 cm]; 91.44 — stopped at 36 in × 2.54 (one yard)*

28–29. Board 7. Rate on Nov 2: $1 = 1.32 CAD; 1 CAD = 16.2 pesos. Theo changes $200 to CAD, then to pesos. How many pesos does he get? → ***4,276.80 pesos*** *3,240 — 200 × 16.2, the first hop skipped [Notes III: the rate's 1 must be the currency you have]; 2,454.55 — divided by 1.32 on the first hop*

30–31. Board 8. 1 km = 0.62 mi. Theo, who weighs 110 lb, rides a 12 km trail. How long is the trail in miles? → ***7.44 mi*** *The 110 lb is not needed. 19.32 — × 1.61 [B1G-M 7.AR.3.3: the inverted ratio]; 49.94 kg — converted the weight instead*

32–33. Board 9. 1 in = 2.54 cm and 1 cm = 0.39 in. Jo converts 30 cm to inches by multiplying by 2.54 and gets 76.2 in. Name the error and fix it. → ***She used the inches ratio on centimeters. Inches are bigger, so fewer: 11.7 in.*** *'She should have used 0.39' with no reason — the fix without the error named; 76.2 defended as 'more inches' — the estimate backwards [B1G-M 7.AR.3.3]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2 take one ratio both ways; 3–4 the next ratio, the second as a vote; 5 chains inside customary then across. The pattern breaks at 6, where three conversions run in a row and the estimate is the only check. 7 is a currency chain, 8 the extra number, 9 the error in writing. The independent set walks the same path: the four length ratios, a chain with an estimate, a currency round trip.

Misconceptions to Watch

Board 1. 7.87 — divided by 2.54 [Notes II: centimeters are smaller, so more of them]

Board 2. 50.8 — multiplied by 2.54 [Notes II: pick the ratio whose 1 is the unit you have]

Board 3. 3.1 — multiplied by 0.62, the other ratio [Notes II]

Board 4. 5 + 0.62: the additive error [7.AR.3.3: Math Nation's named misconception]

Board 5. 15.24 — 6 × 2.54, the feet treated as inches [Notes I: the unit you have must match the ratio's 1]

Board 6. 7.62 — 3 × 2.54, the yards treated as inches [Notes II: estimate first — a 3-yd table is not 8 cm]

Board 7. 3,240 — 200 × 16.2, the first hop skipped [Notes III: the rate's 1 must be the currency you have]

Board 8. 19.32 — × 1.61 [B1G-M 7.AR.3.3: the inverted ratio]

Board 9. 'She should have used 0.39' with no reason — the fix without the error named

Materials Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (they are printed on every item as well). Calculators are allowed on everything and printed on nothing.